Organic Cume 2/15/2013

1. Describe, both with an example and orbital drawings, a Fischer carbene. Give the electron count for the molecule you have shown.

2. Describe, both with an example and orbital drawings, a Schrock carbene or alkylidene complex. Give the electron count for the molecule you have shown.

3. Give the structures for the Grubbs Type I and II catalyst. Discuss whether these are Fischer or Schrock carbenes.

4. Give a catalytic cycle to show the chemistry exhibited by Grubbs catalyst.